

FinRelief AI – AI Powered Debt Relief & Financial Recovery Platform

3. Project Design Phase

System Architecture

User → React.js Frontend → Axios → FastAPI Backend → SQLite Database → Google Gemini AI (with rule-based fallback).

Frontend Design

The frontend is developed using React.js and Vite. It includes Login, Register, Dashboard, Financial Health, Settlement Predictor, Negotiation Email, Know Your Rights, and History pages with reusable components such as Sidebar, Navbar, Cards, and Loan Table.

Backend Design

The backend is implemented using FastAPI. It exposes REST APIs for authentication, financial profile management, loan management, AI services, and history management. JWT authentication secures protected endpoints.

Database Design

SQLite is used as the primary database. Main entities include Users, Financial Profiles, Loans, and History.

API Design

Authentication APIs: Register, Login. Financial APIs: Profile Management. Loan APIs: Add, Update, Delete, View. AI APIs: Financial Health, Settlement Prediction, Negotiation Email, Borrower Rights.

Security Design

Password hashing, JWT authentication, input validation, exception handling, and secure API communication are implemented to improve application security.

User Interface Design

A responsive dark-themed fintech interface provides intuitive navigation, dashboard cards, loan tables, and mobile-friendly layouts.